

Engagement, Performance, and Student Outcomes at the Open University: A Multivariate Analysis

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STA437 — Winter 2026

April 6, 2026

1 Introduction

The Open University Learning Analytics Dataset (OULAD) records six numeric features of assessment behavior for 25,820 student-course registrations: number of prior module attempts, total credits enrolled, mean assessment score, number of assessments submitted, standard deviation of scores, and mean submission date relative to module start [Kuzilek et al., 2017]. Each registration carries one of four outcome labels: Distinction, Fail, Pass, or Withdrawn.

This report asks: *What engagement and performance profiles underlie student outcomes at the Open University, and how well can unsupervised structure in assessment behavior recover the known four-way outcome classification?* Two of the six features (submission count and timing) are observable before any grade is issued, so latent structure built from them could support early-warning systems that must act before module end.

The analysis is a chain of motivated questions. EDA reveals a correlation cluster that motivates a factor model. MVN testing establishes that the data are far from normal, constraining inference. PCA identifies a low-dimensional working space and shows partial class separation, raising whether the latent structure is continuous or discrete. Factor analysis names three dimensions. GMM tests whether those dimensions support discrete student profiles. LDA then tests whether supervised separation uses the same dimensions as unsupervised discovery.

2 Data and Preprocessing

Of the 25,820 registrations, 2,519 (9.8%) have missing `score_std` because standard deviation is undefined for students who submitted exactly one assessment. This missingness is structurally informative: 1,502 of the deleted rows (59.6%) are Withdrawn students and 1,013 (40.2%) are Fail students; no Distinction students appear among them. The complete-case analysis therefore underrepresents the most at-risk registrations, and engagement effect sizes are conservative estimates.

After listwise deletion, $n = 23,301$ complete registrations remain. All six features are standardized before any multivariate analysis. Class proportions are: Pass 53.0%, Fail 20.4%, Withdrawn 13.6%, Distinction 13.0%. A classifier achieving 53.0% adds nothing beyond a majority-class predictor; this is the naive baseline throughout.

3 Exploratory Data Analysis

Figure 1 shows the marginal distributions of all six features and the full correlation matrix. Distinction students occupy the right tail of both `mean_score` and `num_assessments`; Withdrawn students cluster near zero on both. Distributions of `num_of_prev_attempts` and `studied_credits` are discrete and zero-inflated with little class separation.

The correlation matrix reveals a two-block structure. The engagement cluster – `num_assessments` and `avg_submission_date` at $r = 0.651$ – represents students who submit more assessments and submit them later, because additional assessments fall on later deadlines. The score quality cluster – `mean_score` and `score_std` at $r = -0.467$ – represents students whose high average is accompanied by high consistency. A moderate cross-cluster correlation ($r = 0.21$ – 0.28) links the two blocks without merging them. When two variables share 42% of their variance without directly causing each other, the natural generative explanation is a shared latent factor, which motivates the FA in Section 6. But before applying any assumption-dependent method, those assumptions must be tested.

Exploratory Data Analysis: Feature Distributions and Correlation Structure

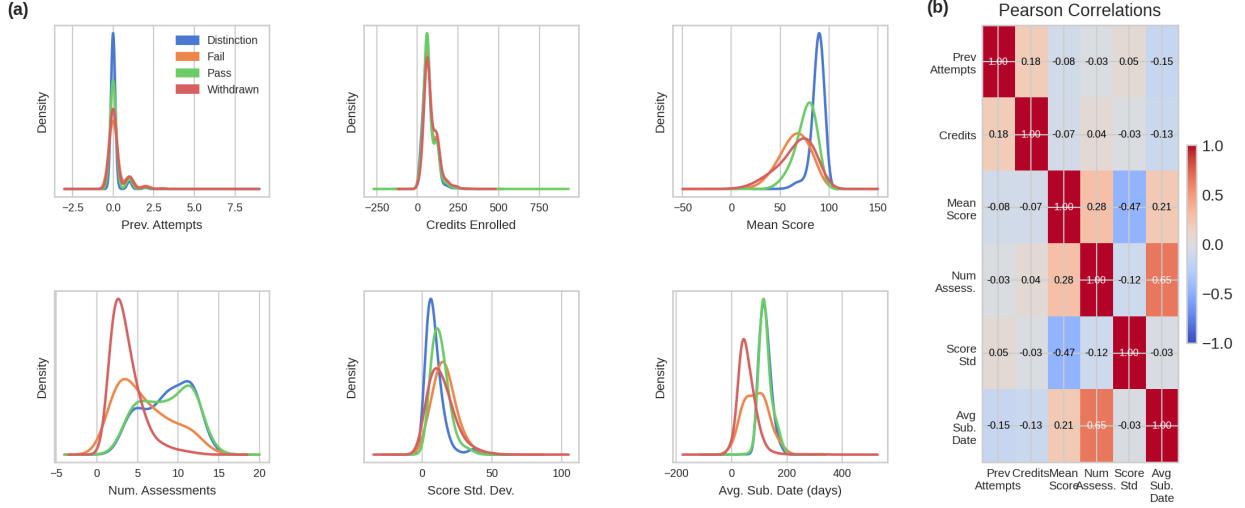


Figure 1: EDA overview. Panel (a): KDEs of each feature by outcome class. Panel (b): Pearson correlation matrix with annotated values. The dominant correlations – `num_assessments` vs. `avg_submission_date` ($r = 0.651$) and `mean_score` vs. `score_std` ($r = -0.467$) – form two partially independent latent clusters. These blocks, not individual features, are the primary structure motivating the factor model in Section 6.

4 Multivariate Normality Assessment

Zero-inflation, discreteness, and skew visible in Figure 1 suggest the six-dimensional distribution is far from multivariate normal (MVN). Establishing this formally constrains which downstream inferences are trustworthy.

Figure 2 plots squared Mahalanobis distances D_i^2 against χ_6^2 quantiles. Under MVN, points follow the diagonal. The upper tail climbs severely above the reference: the maximum $D^2 = 243.4$ versus $\chi_6^2(0.999) = 22.5$, and 6.5% of observations exceed $\chi_6^2(0.975) = 14.4$ (expected: 2.5%). Mardia’s skewness statistic confirms this: $\hat{b}_{1,p} = 20.0$, $\text{stat} = 10,011$, $p \ll 10^{-300}$.

MVN departure does not invalidate PCA or FA as exploratory tools, since neither requires distributional assumptions on the eigendecomposition of the sample correlation matrix. It does require that LDA accuracy be evaluated via cross-validation rather than parametric testing, and it motivates a robust extension: if extreme observations drive the covariance structure, removing them should change predictions materially. Section 9 tests this.

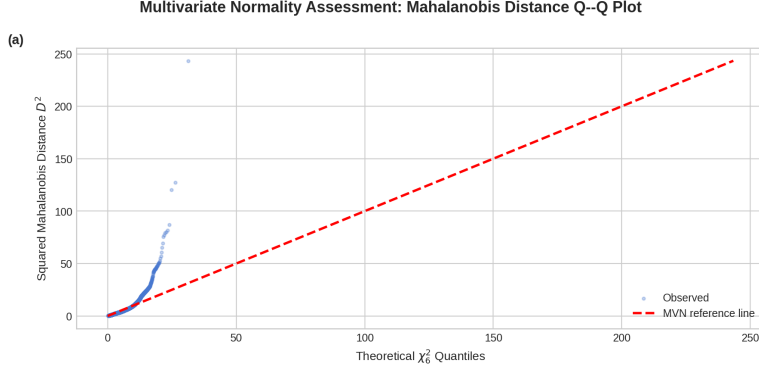


Figure 2: Mahalanobis Q–Q plot. Squared Mahalanobis distances vs. χ_6^2 theoretical quantiles; the diagonal is the MVN reference. The severe upper-tail deviation – maximum $D^2 = 243.4$ against a threshold of 22.5, with 6.5% of points above the 97.5th percentile (expected: 2.5%) – confirms non-normality ($p \ll 10^{-300}$ by Mardia’s test), requiring distribution-free evaluation of all classifiers and motivating the robust LDA check in Section 9.

5 Principal Component Analysis

The EDA correlation blocks suggest the feature space collapses to fewer dimensions. PCA establishes the baseline dimensionality and reveals that engagement and performance are blended onto a single component, motivating a separation that only a factor model can provide.

The scree plot (Figure 3a) shows PC1 and PC2 explaining 32.4% and 20.9% of variance with a clear elbow after PC2; the first three components explain 72.9%. PC1 is dominated by `num_assessments` (+0.560) and `avg_submission_date` (+0.549) with `mean_score` contributing positively (+0.476) – a mixed engagement-and-quality axis. PC2 contrasts `score_std` (+0.670) against `mean_score` (−0.464). The biplot (Figure 3b) shows Distinction students concentrated at high PC1, Withdrawn students near the origin, and Pass/Fail overlapping through the center. This partial separation confirms that PCA directions relate to outcomes, but the overlap and the blended PC1 raise two questions: can the engagement and performance dimensions be cleanly separated, and is the latent structure continuous or discrete?

ICA was considered and rejected: ICA requires statistically independent, super-Gaussian sources. The Q–Q plot shows heavy tails but not strongly super-Gaussian profiles, and ICA components lack the interpretable loading vectors that answer the central question. Kernel PCA was also considered and rejected: Figure 3b shows that the linear separation is already informative, so the added complexity of a nonlinear embedding is not warranted and would sacrifice interpretable loading vectors.

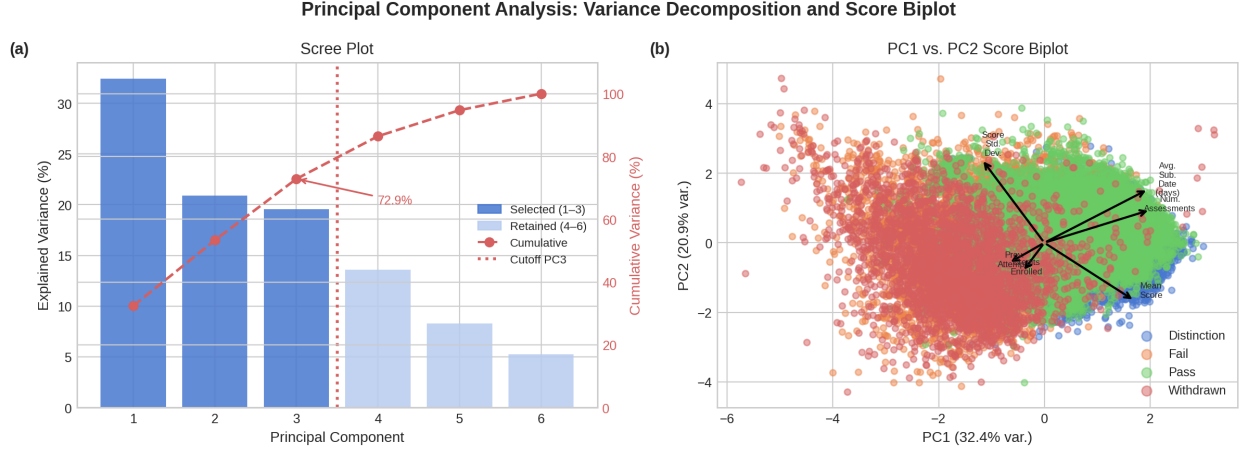


Figure 3: PCA scree and biplot. Panel (a): proportion of variance per component (bars) and cumulative (line); the elbow after PC2 supports a two-dimensional working space at 53.3% explained. Panel (b): PC1 vs. PC2 biplot colored by outcome with loading vectors. Distinction students (blue) concentrate at high PC1; Withdrawn (red) cluster near the origin. The blending of engagement and performance features onto PC1 – and the Pass/Fail overlap in the center – shows that variance-maximizing directions cannot cleanly separate the latent dimensions, directly motivating the factor model.

6 Factor Analysis

PCA showed that engagement and performance are blended onto PC1. FA asks whether the correlation block structure is better explained by named latent factors – each causing the within-block correlations – and whether those factors can be separated cleanly.

Two pre-tests validate the approach. KMO is 0.524, meeting the minimum threshold of 0.5 (`num_of_prev_attempts` and `studied_credits` are borderline contributors and their factor loadings should be interpreted with more caution). Bartlett’s test rejects the identity-matrix null ($\chi^2_{15} = 23,413$, $p \ll 10^{-300}$). ICA was considered as an alternative to FA and rejected: ICA requires statistically independent, super-Gaussian sources and defines components by statistical properties rather than interpretable loadings. FA’s diagonal noise model, which separates shared-factor variance from feature-specific residuals, is the correct model for the block correlation structure identified in EDA.

Parallel analysis retained three factors: actual eigenvalues 1.956, 1.258, and 1.160 each exceed the corresponding 95th-percentile thresholds from 500 random matrices (1.066, 1.038, 1.019); the fourth (0.829) falls below its threshold. A two-factor solution was examined and rejected: it merged Consistent Performance and Prior Commitment into a single less-interpretable factor, losing the distinction between current-module score quality and prior academic investment.

The three-factor solution used principal-axis factoring on the full correlation matrix with Kaiser varimax rotation. Varimax is appropriate because the EDA block structure implies each feature should load primarily on one factor (simple structure). Oblique rotation was not used because the cross-cluster correlation ($r = 0.21\text{--}0.28$) does not motivate correlated factors. Loadings are displayed in Figure 4 and Table 1.

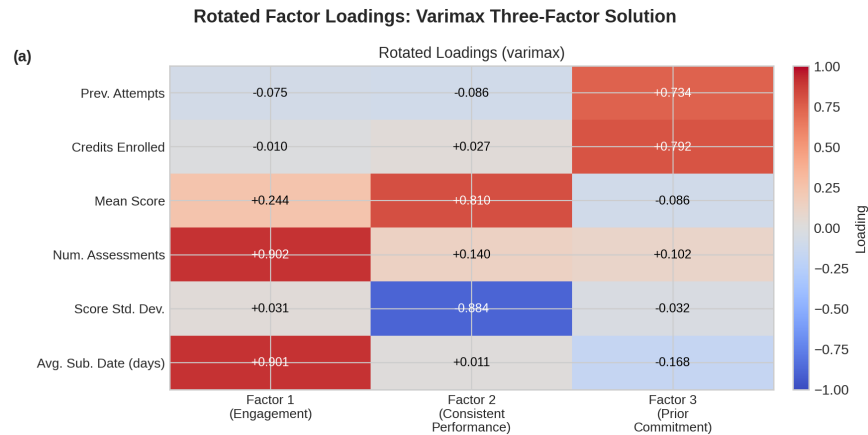


Figure 4: Varimax factor loading heatmap. Loadings of each feature on the three retained factors; red is positive, blue is negative. Factor 1 (Engagement) loads on submission count and timing (> 0.90); Factor 2 (Consistent Performance) contrasts mean score against score variability; Factor 3 (Prior Commitment) captures prior attempts and credits. The clean simple structure confirms that the two EDA correlation blocks reflect genuinely separable latent dimensions, setting up the convergence test with LDA in Section 8.

Table 1: Varimax-rotated factor loadings (three factors). h^2 is communality (sum of squared loadings per feature). Each feature loads strongly on exactly one factor, confirming simple structure. The Engagement factor (F1) comprises features observable before any grade is issued – submission count and timing – making it the basis for early outcome prediction examined in Section 8.

Feature	F1 Engagement	F2 Consist. Perf.	F3 Prior Commit.	h^2
Prev. Attempts	−0.075	−0.086	+0.734	0.55
Credits Enrolled	−0.010	+0.027	+0.792	0.63
Mean Score	+0.244	+0.810	−0.086	0.72
Num. Assessments	+0.902	+0.140	+0.102	0.84
Score Std. Dev.	+0.031	−0.884	−0.032	0.78
Avg. Sub. Date (days)	+0.901	+0.011	−0.168	0.84
Var. explained (%)	28.2	24.4	20.2	72.8

Factor 1 (Engagement) loads on `num_assessments` and `avg_submission_date` (+0.902 and +0.901), the pair forming the $r = 0.651$ EDA correlation cluster. This latent quantity – persistent engagement across the module – causes students who submit more to also submit later. Factor 2 (Consistent Performance) contrasts high `mean_score` (+0.810)

against high `score_std` (-0.884), identifying the $r = -0.467$ quality cluster: a student high on F2 scores consistently, not merely highly. Factor 3 (Prior Commitment) captures `num_of_prev_attempts` ($+0.734$) and `studied_credits` ($+0.792$), features orthogonal to the assessment-derived clusters. The FA finding raises a natural next question: does this continuous three-dimensional structure also support discrete student profiles?

7 Gaussian Mixture Modeling

FA established continuous latent structure. GMM tests whether students also group into discrete profiles by fitting a probabilistic mixture of Gaussians and comparing fit across K .

GMM was run in the two-dimensional PCA score space (53.3% of variance) rather than full six-dimensional space. This removes residual noise attributed to components 3–6 by the factor model, stabilizes full-covariance estimation, and allows visual validation against the Figure 3b biplot. k-means was not used because it requires spherical, equal-size clusters; the imbalanced outcome classes and elongated point clouds in the biplot violate both. The Factor 3 (Prior Commitment) features have near-zero loadings on PC1 and PC2, so the GMM clusters reflect Factors 1 and 2 only; this does not diminish Factor 3’s existence, only confirms it operates on a dimension orthogonal to the engagement- performance space.

BIC decreases monotonically from 153,075 ($K = 1$) to 145,883 ($K = 6$). The marginal improvement from $K = 1$ to $K = 4$ is 6,193 across three components; from $K = 4$ to $K = 6$ it is only 999 across two more. Inspection of the $K = 5$ and $K = 6$ solutions shows the additional components subdivide the large Pass cluster without producing new modal outcome labels. $K = 4$ is selected as the point where marginal gains become small and cluster labels remain interpretable.

The four clusters (Figure 5b): Cluster 3 ($n = 8,280$; 68% Pass / 27% Distinction) occupies the high-engagement region; Cluster 1 ($n = 5,820$; 70% Pass) is moderate-engagement; Cluster 4 ($n = 3,049$; 43% Fail / 31% Withdrawn) is the disengagement cluster; Cluster 2 ($n = 6,152$) spans the transition zone. The adjusted Rand index comparing cluster assignments to outcome labels is 0.137, where 0 is chance-level agreement and 1 is perfect recovery. The modest ARI directly answers the second half of the central question: assessment behavior partially recovers the outcome taxonomy but cannot fully recover it, because Distinction and Pass students share the high-engagement cluster – what separates them (grading thresholds, examination performance) is not in these six features.

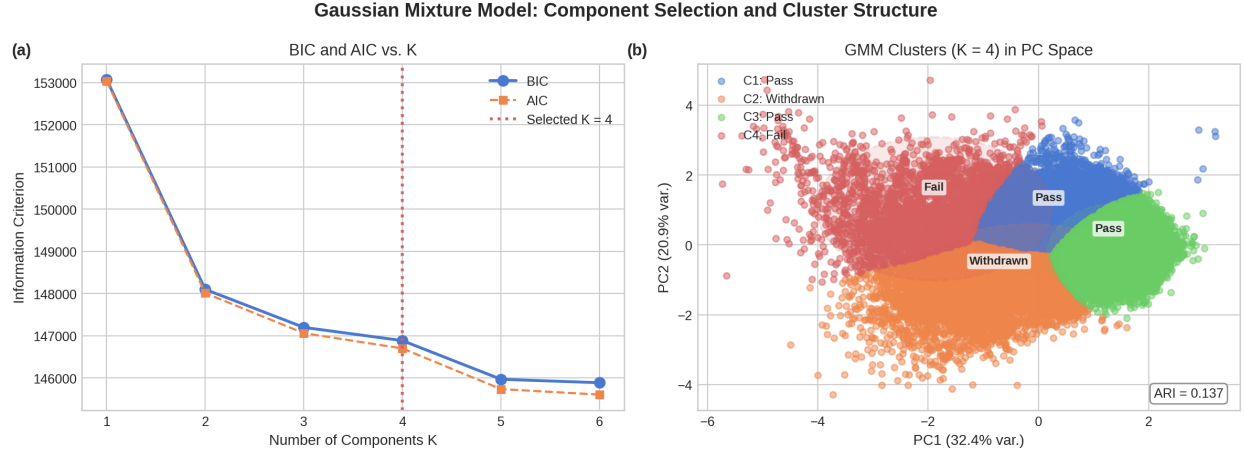


Figure 5: GMM component selection and cluster structure. Panel (a): BIC and AIC vs. K ; the vertical line marks the selected $K = 4$, where marginal BIC gains decline sharply and additional components merely subdivide the Pass cluster. Panel (b): PC1-PC2 scatter with 2σ ellipses and modal outcome labels per cluster. ARI = 0.137 (0 = chance, 1 = perfect) confirms that unsupervised assessment-behavior structure partially recovers the outcome taxonomy – the quantitative answer to the second part of the central question.

8 Linear Discriminant Analysis

GMM found that outcome groups differ in the feature space but that assessment behavior cannot fully recover the label boundaries. LDA tests whether supervised separation relies on the same latent dimensions as unsupervised discovery – the convergence test that completes the analysis.

LDA was fitted on all six standardized features with four outcome classes. LD1 captures 89.2% of between-class variance. Its dominant loadings are `avg_submission_date` (+0.928) and `mean_score` (+0.597), with `num_assessments` contributing positively (+0.287) and the Factor 3 features (`num_of_prev_attempts`, `studied_credits`) near zero (−0.030 and −0.026). This mirrors the FA result precisely: the two features with the highest communalities in the Engagement ($h^2 = 0.84$) and Consistent Performance ($h^2 = 0.72$) factors are also the dominant LD1 contributors, while Factor 3 features carry no discriminatory weight. Three independent methods – FA, GMM, and LDA – identify the same two dimensions. Prior Commitment is a real latent dimension but does not predict current-module outcomes.

The LD1 histogram (Figure 6) shows Distinction students at the high end, Withdrawn students at the low end, and Pass/Fail overlapping in the center. Five-fold CV accuracy is 64.5% ($\pm 0.4\%$), 11.5 points above the 53.0% naive baseline. Per-class recall reveals the asymmetry: Pass 94.4%, Withdrawn 63.2%, Distinction 3.2%. LDA effectively screens for withdrawal risk but conflates Distinction with Pass – the same boundary GMM could not

recover, providing convergent evidence.

LDA’s shared-covariance assumption is violated given confirmed non-normality. QDA would relax it but introduces estimation variance for the small Distinction class ($n = 3,024$, 13%); cross-validation provides an honest, assumption-free accuracy estimate regardless of the violation.

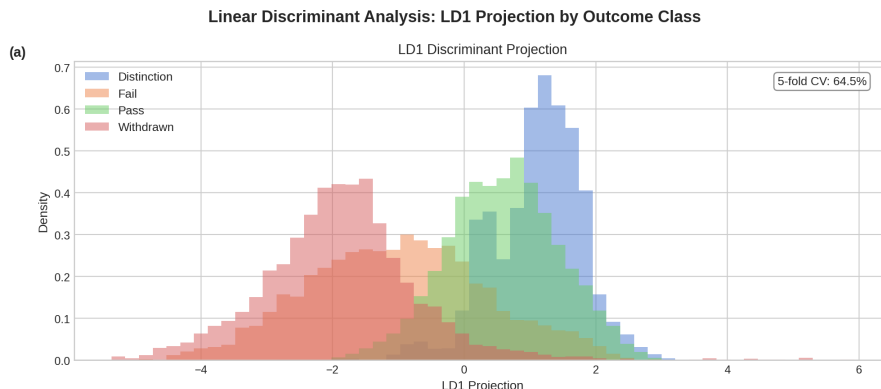


Figure 6: LD1 projection by outcome class. Normalized density histograms for each outcome along the first linear discriminant. LD1 is dominated by `avg_submission_date` and `mean_score`, producing a gradient from Withdrawn (left) to Distinction (right). CV accuracy of 64.5% is 11.5 points above the naive baseline; per-class recall (Pass 94.4%, Withdrawn 63.2%, Distinction 3.2%) shows LDA is useful for withdrawal screening but cannot recover the Distinction/Pass boundary – converging with the GMM ARI finding.

9 Extensions: Comparison and Robustness

Logistic regression. Multinomial logistic regression was fitted as an alternative classifier. LR relaxes LDA’s shared-covariance assumption while remaining linear, keeping the LDA/LR comparison clean (both fit linear boundaries) and avoiding the estimation-variance problem that makes QDA unattractive for the small Distinction class. LR achieves 67.0% ($\pm 0.8\%$) CV accuracy versus LDA’s 64.5%. The gain is concentrated on Distinction recall, which rises from 3.2% (LDA) to 39.3% (LR), confirming that the shared-covariance assumption is most severely violated for the class with the most distinctive feature profile.

Robust LDA. The MVN assessment identified 6.5% of observations above $\chi^2_6(0.975)$. Per-fold robust LDA was fitted by removing these outliers from the training data before fitting. Robust LDA achieves 65.1% ($\pm 0.6\%$), statistically indistinguishable from standard LDA’s 64.5% (Figure 7). Outliers inflate variance estimates but do not shift class means materially; the discriminant direction is stable.

Missing data robustness. The full FA was re-run after imputing `score_std = 0` for

single-assessment students. Engagement factor loadings change by at most 0.01; Consistent Performance loadings by at most 0.03. The qualitative conclusions are stable under imputation.

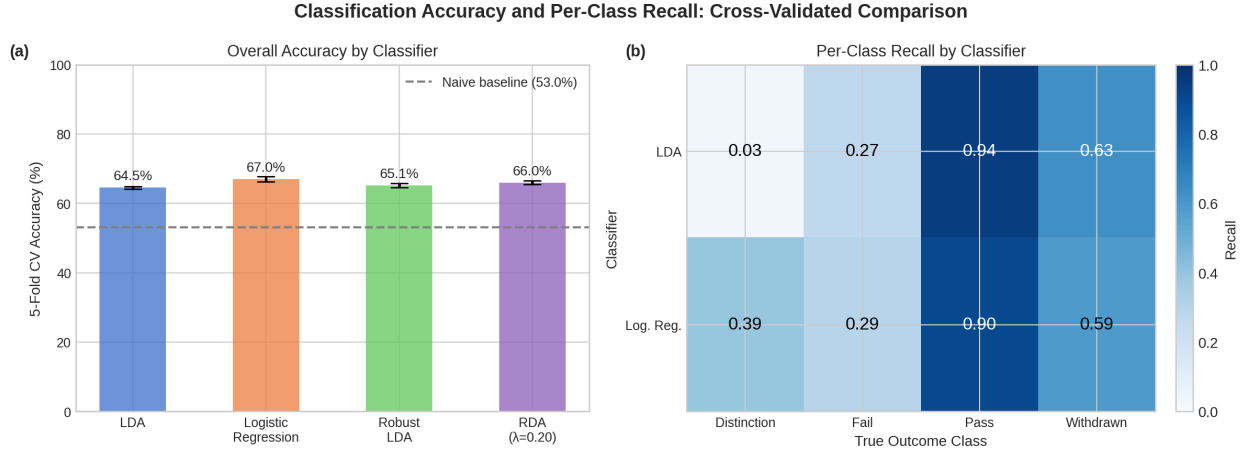


Figure 7: Classifier comparison and per-class recall. Panel (a): five-fold CV accuracy for LDA (64.5%), logistic regression (67.0%), and robust LDA (65.1%); the dashed line is the 53.0% naive baseline. Panel (b): per-class recall heatmap. Logistic regression substantially outperforms LDA on Distinction recall (39.3% vs. 3.2%), showing the shared-covariance assumption is most violated for the minority high-performing class. Robust LDA stability confirms that the 6.5% of outlying observations do not drive the discriminant direction.

10 Synthesis and Conclusion

Three independent methods converge on the same finding. FA identified an Engagement factor (submission count and timing) and a Consistent Performance factor (high mean score, low score variability). GMM placed high-engagement students in Pass/Distinction clusters and low-engagement students in a Fail/Withdrawn cluster. LDA confirmed that LD1 is dominated by the highest- communality features of the Engagement ($h^2 = 0.84$) and Consistent Performance ($h^2 = 0.72$) factors, while Factor 3 (Prior Commitment) has near-zero LD1 loadings and does not drive GMM cluster separation. Prior Commitment describes a student’s academic history but not their current- module outcomes.

A divergent finding separates the methods. LDA recovers Withdrawn students with 63.2% recall but Fail students with only 26.7%; GMM places both in the same low-engagement cluster. Withdrawn students have a more extreme low-engagement profile than Fail students – total rather than partial disengagement – which LDA captures through the submission-date loading but GMM cannot resolve within a shared cluster. This distinction matters practically: Engagement supports an early-withdrawal screen built on submission timing, while Consistent

Performance supports a quality-risk screen built on score trajectories.

The central limitation is scope. The analysis excludes the 6,773 students who never submitted, and the 2,519 deleted rows are 60% Withdrawn. Effect sizes are conservative; the never-submitted group – the highest withdrawal risk – remains entirely outside this analysis. The robustness checks confirm that including single-assessment students via imputation does not change the qualitative conclusions, but the fundamental gap remains.

References

Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics dataset. *Scientific Data*, 4, 170171. <https://doi.org/10.1038/sdata.2017.171>